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Overview

The Dock Motor Speed Procedure is required as part of setup and calibration. Procedures for opening magnet enclosure covers and adjusting patient dock hardware (Motor On Switch Actuation Adjustment and Dock Motor Gear Assembly Adjustments) are provided in the event that related dock problems arise.

1- DOCK MOTOR SPEED ADJUSTMENT

Motor speed is affected by magnetic field effects on the motor and the moving parts in the patient transport. The dock motor speed cannot be factory preset; it must be adjusted for each individual site. The motor speed is set for an unloaded condition that equates to 360 rpm (or six revolutions per second). The adjustment can be made using the preferred *Section 2-2 Two-Man Procedure* or, when short handed, *Section 2-3 One-Man Procedure*. A DVM is required to verify power is off. The MEPS measurement for the Dock Motor Speed Adjustment requires one oscilloscope probe. The SSM differential measurement for the Dock Motor Speed Adjustment requires two oscilloscope probes because the Power Supply circuitry is floating.

1-1 Two-Man Procedure



The system must be setup for a Body mode protocol and the [Prepare to Scan] softkey selected initially. Pulse Body mode once at TG of 0. Do NOT open the SSM top cover for 1 minute after the system has transitioned from Head, Surface, M/C or MNS mode to Body mode. Failure to follow these instructions resulted in the Dynamic Disable IGBT Circuitry (Direct Drive, Body1, Body2) overheating and shorting requiring CPD replacement.

1. Set up a Body mode protocol and select **[Prepare to Scan]** softkey. Wait 5 minutes before opening the SSM cover. (See above Caution for detail.)
2. Dock the patient transport, and remove the head coil carriage from the patient table if necessary. Lower table to fully down position using the foot pedals at either end of the patient transport.
3. Remove power to the dock. Refer to the Safety procedure, *Section 6 - OSHA Lock-Out/Tag-Out Requirements* (SYSMSA2.DOC).
4. Lock-Out / Tag-Out circuit breaker MR1 A14 CB6 at rear of **RF/Pen Cabinet**, or circuit breaker MR1 A20 CB4 at rear of SSM in **RF/Pen II** and **RF/PDU Cabinet**.
5. Verify that voltage between Dock and Dock RTN is **0 VDC** using a Digital Volt Meter (DVM):

RF/Pen Cabinet: Locate the MEPS Module at the bottom of the Cabinet and the test point strip at the center of the rear panel of the MEPS module, and connect the DVM.

RF/Pen II and RF/PDU Cabinet: Locate the SSM at the bottom of the Cabinet and the test point strip at the center of the rear panel of the SSM, and connect the DVM.

6. Remove the four screws and four No. 10 flat washers securing the MEPS or SSM module to the cabinet.

7. Slide the MEPS or SSM module out the front of the cabinet.

RF/Pen Cabinet:

- a. Loosen the three 4-40 captive screws on the top cover of the MEPS module. Open and remove the lid. Locate the High Current power supply board that contains the voltage potentiometer R38.
- b. Pull out the RFSC and loosen the captive screws on the top cover of the SSM to gain access to the white plunger switch. Open the lid. Pull the High Voltage enable plunger switch to the **Up** position.
- c. Connect Channel 1 oscilloscope lead to TP3 (Dock 38.5V), and connect Channel 1 oscilloscope ground reference lead to TP1 or TP4 or TP7 (SGND).
- d. Set Channel 1 trace to zero reference graticle on the oscilloscope display. Adjust time/div to approximately **100 mS/div**. Set oscilloscope channels to **AC Coupling**.

RF/Pen II Cabinet and RF/PDU Cabinet:

- a. Do **not** disable the HV at the front of the SSM.
- b. Remove the twelve 4-40 screws on the top cover of the SSM. Open the lid, and locate the Dock/Light power supply board that contains the voltage potentiometer R80.
- c. Connect Channel 1 oscilloscope lead to TP3 (38.5V, Dock), connect Channel 2 oscilloscope lead to TP4 (DOCK-RTN).
- d. Connect both oscilloscope lead grounds together. Isolate ground leads with tape as required to ensure no contact with electrical circuitry.
- e. Set both channels to ground reference, and set Channel 2 to **add/invert**. Adjust both ground leads to the same zero reference graticle on the oscilloscope display. Adjust time/div to approximately **100 mS/div**. Set oscilloscope channels to **AC Coupling**.



SHOCK HAZARD! THE SSM IPS ASSEMBLY CIRCUITRY IS FLOATING. DO NOT REFERENCE TEST EQUIPMENT TO GROUND CHASSIS OR ANY GROUND TEST POINTS.

8. Lower Table to the full down position using the Table Down pedal.
9. Switch on circuit breaker MR1 A14 CB6 on the **RF/Pen Cabinet**, or switch on circuit breaker MR1 A20 CB4 on the **RF/Pen II** and **RF/PDU Cabinet**.

Note

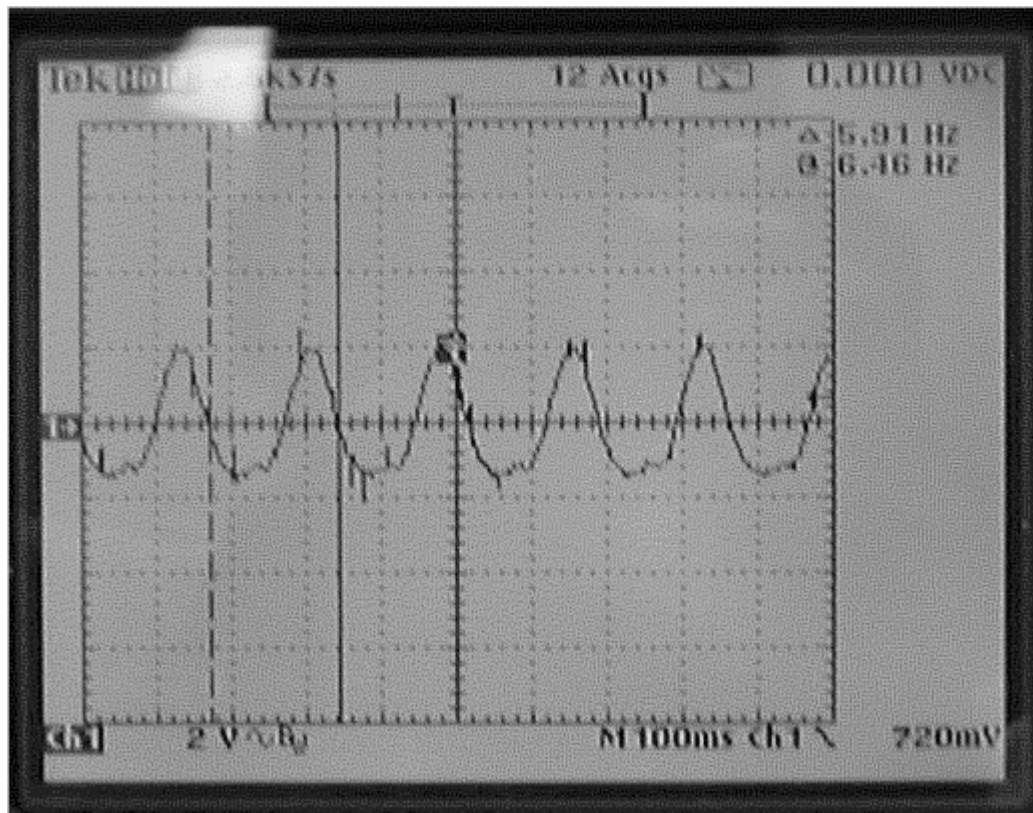
Only perform this procedure with the system set up for a Body mode protocol, or the Direct Drive IGBT circuitry may short.

10. Monitor output at test points while a second person presses the Table Up pedal on the dock, raising the table. (Refer to the Caution at the beginning of this section for details.)

Note

This measurement is valid only during the period of table top upward motion. Once the table top reaches the maximum height, dock motor will be too loaded to provide accurate adjustment measurement.

11. Adjust R38 at the **RF/Pen Cabinet** and R80 at the **RF/Pen II Cabinet** and **RF/PDU Cabinet** to obtain a voltage waveform of six pulses per second, $\pm 1/4$ pulse per second. (Illustration 1-1 shows 5 and 3/4 pulses/second.) Refer to above Note.



6 PULSES PER SECOND OSCILLOSCOPE DISPLAY
ILLUSTRATION 1-1

12. Repeat this procedure as necessary or whenever the Dock circuit board is replaced.

1-2 One-Man Procedure



The system must be setup for a Body mode protocol and the [Prepare to Scan] softkey selected initially. Pulse Body mode once at TG of 0. Do NOT open the SSM top cover for 1 minute after the system has transitioned from Head, Surface, M/C or MNS mode to Body mode. Failure to follow these instructions resulted in the Dynamic Disable IGBT Circuitry (Direct Drive, Body1, Body2) overheating and shorting requiring CPD replacement.

1. Set up a Body mode protocol and select [Prepare to Scan] softkey. Wait 5 minutes before opening the SSM cover. (Refer to above Caution for detail.)
2. Remove power to dock. Refer to the Safety procedure, *Section 6 - OSHA Lock-Out/Tag-Out Requirements* (SYSMSA2.DOC).
3. Lock-Out / Tag-Out circuit breaker MR1 A14 CB6 at rear of **RF/Pen Cabinet**, or circuit breaker MR1 A20 CB4 at rear of SSM in the **RF/Pen II** and **RF/PDU Cabinet**.
4. Verify that voltage between Dock and Dock RTN is **0 VDC** using a DVM:

RF/Pen Cabinet: Locate the MEPS Module at the bottom of the Cabinet, and locate the test point strip at the center of the rear panel of the MEPS module, and connect the DVM.

RF/Pen II and RF/PDU Cabinet: Locate the SSM at the bottom of the Cabinet and the test point strip at the center of the rear panel of the SSM. Connect the DVM.

5. Remove the four screws and four No. 10 flat washers securing the MEPS or SSM module to the cabinet.
6. Slide the MEPS or SSM module out from the front of the cabinet.

RF/Pen Cabinet:

- a. Loosen the three 4-40 captive screws on the top cover of the MEPS module. Open and remove the lid. Locate the High Current power supply board that contains the voltage potentiometer R38.
- b. Pull out the RFSC and loosen the captive screws on the top cover of the SSM to gain access to the white plunger switch. Open the lid. Pull the High Voltage enable plunger switch to the **Up** position.
- c. Connect Channel 1 oscilloscope lead to TP3 (Dock 38.5V), and connect Channel 1 oscilloscope ground reference lead to TP1 or TP4 or TP7 (SGND).
- d. Set Channel 1 trace to zero reference graticle on the oscilloscope display. Adjust time/div to approximately **100 mS/div**. Set oscilloscope channels to **AC Coupling**.

RF/Pen II Cabinet and RF/PDU Cabinet:

- a. Do not disable the HV at the front of the SSM.
- b. Remove the twelve 4-40 screws on the top cover of the SSM. Open the lid, and locate the Dock/Light power supply board that contains the voltage potentiometer R80.
- c. Connect Channel 1 oscilloscope lead to TP3 (38.5V, Dock), and connect Channel 2 oscilloscope lead to TP4 (DOCK-RTN)
- d. Connect both oscilloscope lead grounds together. Isolate ground leads with tape as required to ensure no contact with electrical circuitry.
- e. Set both channels to ground reference, and set Channel 2 to **add/invert**. Adjust both ground leads to the same zero reference graticle on the oscilloscope display. Adjust time/div to approximately **100 mS/div**. Set oscilloscope channels to **AC Coupling**.



SHOCK HAZARD! THE SSM IPS (INTEGRATED POWER SUPPLY) ASSEMBLY CIRCUITRY IS FLOATING. DO NOT REFERENCE TEST EQUIPMENT TO GROUND CHASSIS OR ANY GROUND TEST POINTS.

7. Lower Table to the full down position using the Table Down pedal.
8. Place a weight on the Table Up and Table Down foot pedals on the Dock. (The weighting of both pedals will ensure that the table does not top out and pull more current than normal.)
9. Switch on circuit breaker MR1 A14 CB6 in the **RF/Pen Cabinet** or circuit breaker MR1 A20 CB4 in the **RF/Pen II** and **RF/PDU Cabinet**.

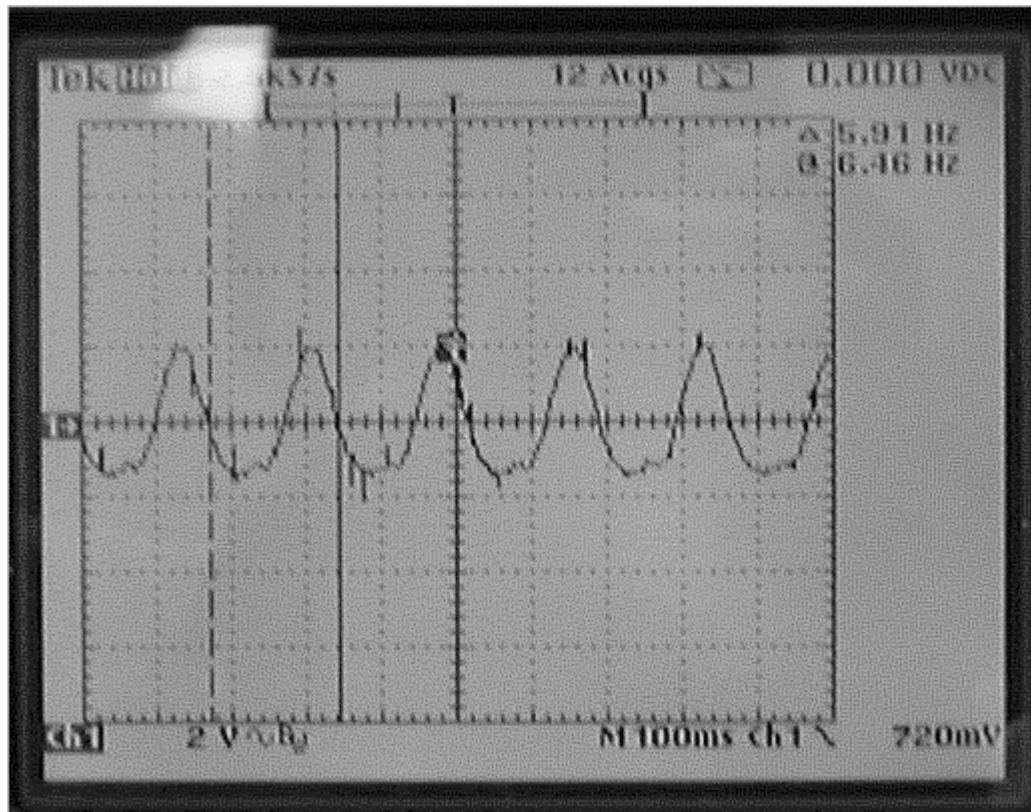
Note

Only perform this procedure with the system set up for a Body mode protocol. (Refer to the Caution at the beginning of this section for details.)

10. Monitor output at test points (The following test setup measurement is valid only for approximately 12 seconds):

RF/Pen Cabinet: Adjust R38 to obtain a voltage waveform of six pulses per second, $\pm 1/4$ pulse per second. (Illustration 1-2 shows 5 and 3/4 pulses/second.)

RF/Pen II Cabinet and RF/PDU Cabinet: Adjust R80 to obtain a voltage waveform of six pulses per second, $\pm 1/4$ pulse per second. (Illustration 1-2 shows 5 and 3/4 pulses/second.)

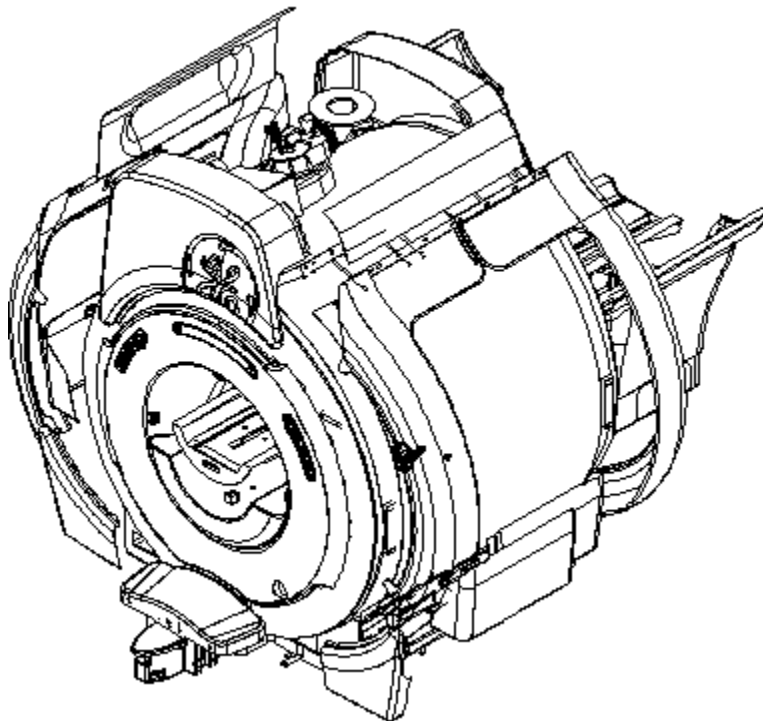


6 PULSES PER SECOND OSCILLOSCOPE DISPLAY ILLUSTRATION 1-2

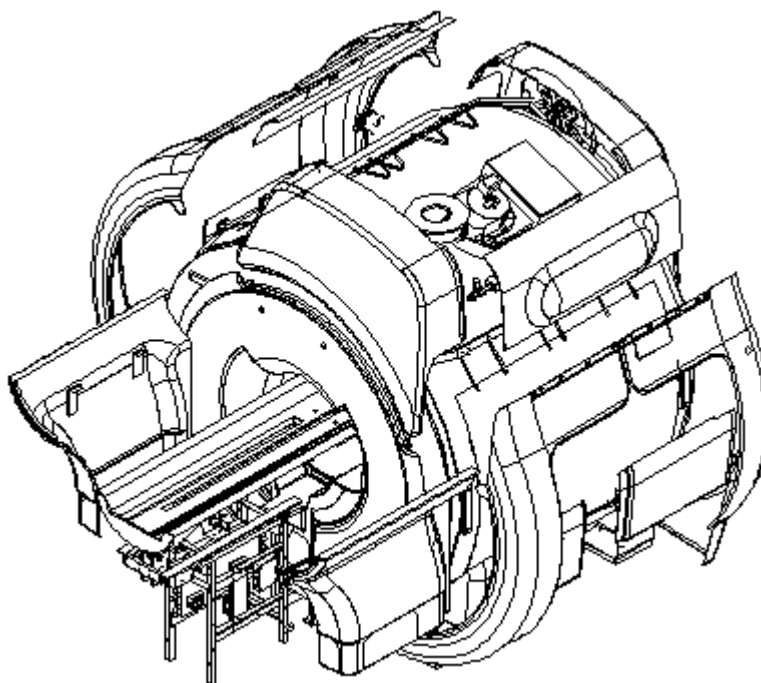
11. Repeat this procedure as necessary or whenever the Dock circuit board is replaced.

2- WIDE OPEN MAGNET ENCLOSURE (MG2)

The procedures that follow are for opening magnet enclosure covers and adjusting patient dock hardware. The magnet wide open enclosure consists of 18 panels (all removable), plus the front and rear endbells. (See Illustrations 2-1 and 2-2.)



ACCESS TO MAGNET ENCLOSURE - FRONT VIEW
ILLUSTRATION 2-1



ACCESS TO MAGNET ENCLOSURE - REAR VIEW
ILLUSTRATION 2-2

Note

All enclosure component seams and gaps, visible to the operator and patient, shall be less than 0.25 inches (6.35 mm). Exceptions may be made in areas where esthetic visual enhancement of the component gap is required by industrial design.

- Gap at the seam between the enclosure and RF coil shall **not** exceed 0.125 inches (3.175 mm).
- Gap uniformity along any continuous seam shall be ± 0.0625 inch (1.59 mm).

3- MOTOR ON SWITCH ACTUATION ADJUSTMENT

3-1 Description

The Motor On Actuation Switch in the dock assembly should be adjusted so it is activated when the Table Up foot pedal is pressed approximately 3/4 of its total length of travel. This is factory adjusted. However, it is possible to adjust the position of the Motor On Switch in the field to maintain the proper actuation level.



FERROUS MATERIAL HAZARD! THE DOCK ASSEMBLY MAIN DOCK COVER CONTAINS A MOTOR MADE WITH FERROUS COMPONENTS. HOLDING THIS ASSEMBLY TOO CLOSE TO THE MAGNET BORE WILL CAUSE IT TO BE FORCIBLY ATTRACTED TO THE MAGNET. TO PREVENT POSSIBLE BODILY INJURY OR DAMAGE TO MOTOR, MAGNET ENCLOSURE OR MAGNET, PLACE DOCK ASSEMBLY ON FLOOR AND SLIDE IT OUT OF EXAM ROOM.

3-2 Procedure

1. Remove power to dock. Refer to the Safety procedure, *Section 6 - OSHA Lock-Out/Tag-Out Requirements (SYSMSA2.DOC)*.

A graphic of a warning label with the word "WARNING!" in a bold, sans-serif font, enclosed within a rectangular box with a 3D effect.

STRONG MAGNETIC FIELD! WHEN SERVICING ANY MAGNETIC EQUIPMENT, IT IS CRITICALLY IMPORTANT THAT THE SERVICE ENGINEER CONSCIOUSLY PLAN THE PATH TO BE TAKEN WHEN MOVING HIGHLY FERROUS DEVICES IN THE MAGNET ENVIRONMENT. THE PATH SHOULD BE AS FAR FROM THE MAGNET AS PRACTICAL.

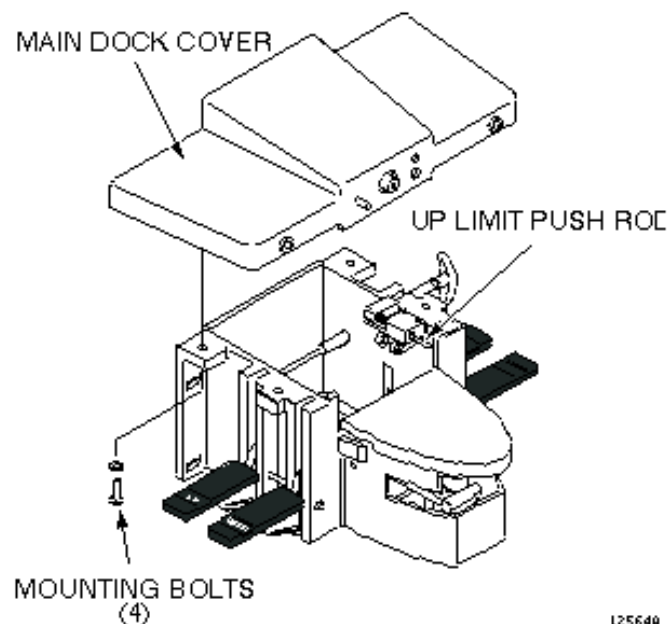
SAFETY REQUIREMENTS:

- **THE SERVICE PATH MUST ALLOW FOR THE DOCK TO BE MOVED ALONG THE GROUND A DISTANCE EQUAL TO THAT OF THE DISTANCE BETWEEN THE MAGNET AND THE FOOT-END OF THE PATIENT TABLE PRIOR TO BEING LIFTED VERTICALLY.**

- **TWO (2) MR SAFETY TRAINED PERSONNEL MUST BE PRESENT AT ALL TIMES WHEN SERVICING HIGHLY FERROUS DEVICES IN THE AREAS OF MAGNETIC FIELDS.**

WHEN PLANNING A SERVICE PATH, IT IS CRITICAL THAT THE PATH BE CLEAR AND SUFFICIENTLY WIDE. ENSURE THAT THERE ARE NO TRIP HAZARDS, OBSTACLES, CLUTTER, SLIPPERY SURFACES OR OTHER ITEMS EVEN PARTIALLY RESTRICTING THE PATH. IF THERE ARE PORTABLE OBSTACLES IN A PATH, REMOVE THEM FROM THE AREA AND REPLACE THEM AFTER THE SERVICE ACTION IS COMPLETED. IT IS REQUIRED TO WALK THE PATH PRIOR TO BEGINNING SERVICE TO ENSURE THAT THERE IS SUFFICIENT SPACE THROUGH WHICH TO PASS FOR YOURSELF AND THE OBJECT BEING SERVICED.

2. Remove dock from the Magnet Room following instructions in *Section 3-1-1 Dock Assembly Removal* in *SRI and Dock Hardware* (ME1REA1.PDF).
3. Once dock is removed from the Magnet Room, remove the four mounting bolts from the main dock cover shown in Illustration 3-1.

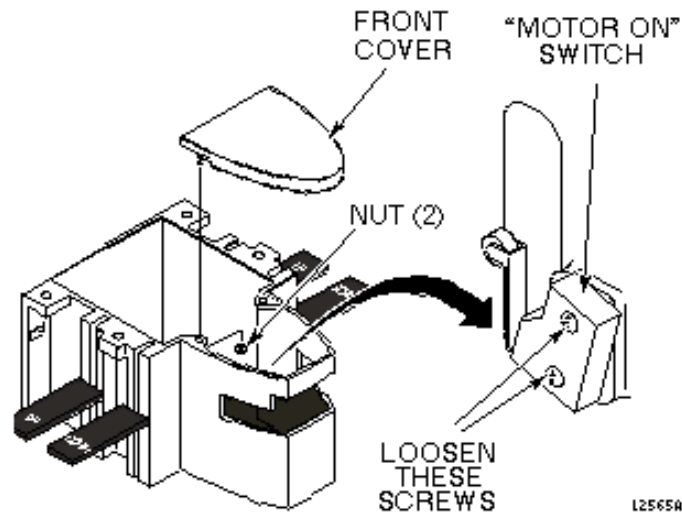


DOCK COVER REMOVAL
ILLUSTRATION 3-1

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4. Slide main dock cover forward until the Up Limit Switch pushrod just clears the main dock cover holes (Illustration 3-1).

5. Remove front cover by removing two nuts (shown in Illustration 3-2) securing front cover to the dock.



MOTOR ON SWITCH ADJUSTMENT
ILLUSTRATION 3-2

6. Loosen two mounting screws securing Motor On Switch to side of dock assembly. (See Illustration 3-2.)
7. Adjust the Motor On Switch position so it is activated (clicks) when the Table Up foot pedal is pressed approximately 3/4 of its total length of travel.
8. After adjustment is complete, tighten the two mounting screws.
9. Reinstall front cover by attaching the two mounting nuts.
10. Carefully position the main dock cover so that Up Limit Switch pushrod is aligned with upper hole in main dock cover.
11. Replace main dock cover by securing the four mounting bolts shown in Illustration 3-1.



FERROUS MATERIAL HAZARD! THE DOCK ASSEMBLY MAIN DOCK COVER CONTAINS A MOTOR MADE WITH FERROUS COMPONENTS. HOLDING THIS ASSEMBLY TOO CLOSE TO THE MAGNET BORE WILL CAUSE IT TO BE FORCIBLY ATTRACTED TO THE MAGNET. TO PREVENT POSSIBLE BODILY INJURY OR DAMAGE TO MOTOR, MAGNET ENCLOSURE OR MAGNET, PLACE DOCK ASSEMBLY ON FLOOR AND SLIDE IT OUT OF EXAM ROOM.

12. Reinstall dock in the Magnet Room following *Section 3-1-2 Dock Assembly Installation* in *SRI and Dock Hardware* (ME1REA1.PDF).
13. Ensure that dock Motor On Switch operates properly when Table Up pedal is pressed.

4- DOCK MOTOR GEAR ASSEMBLY ADJUSTMENT

The dock motor gear box has a clutch that prevents backlash on the gears in the gear box. This clutch is not intended to regulate dock motor speed. (Refer to *Section 1 Dock Motor Speed Adjustment*.) This clutch simply prevents gear backlash that would make the gear assembly noisy.

There are three probable causes of a noisy gear assembly:

1. Cam improperly adjusted
2. Excessive play on motor tube spring
3. Clutch improperly adjusted

Perform procedures in Sections 4-2, 4-3 and 4-4 in the sequence presented to correct excessive gear assembly noise.

4-1 Cam Adjustment

1. Remove power to the dock. Refer to the Safety procedure, *Section 6 - OSHA Lock-Out/Tag-Out Requirements* (SYSMSA2.DOC).

WARNING!

STRONG MAGNETIC FIELD! WHEN SERVICING ANY MAGNETIC EQUIPMENT, IT IS CRITICALLY IMPORTANT THAT THE SERVICE ENGINEER CONSCIOUSLY PLAN THE PATH TO BE TAKEN WHEN MOVING HIGHLY FERROUS DEVICES IN THE MAGNET ENVIRONMENT. THE PATH SHOULD BE AS FAR FROM THE MAGNET AS PRACTICAL.

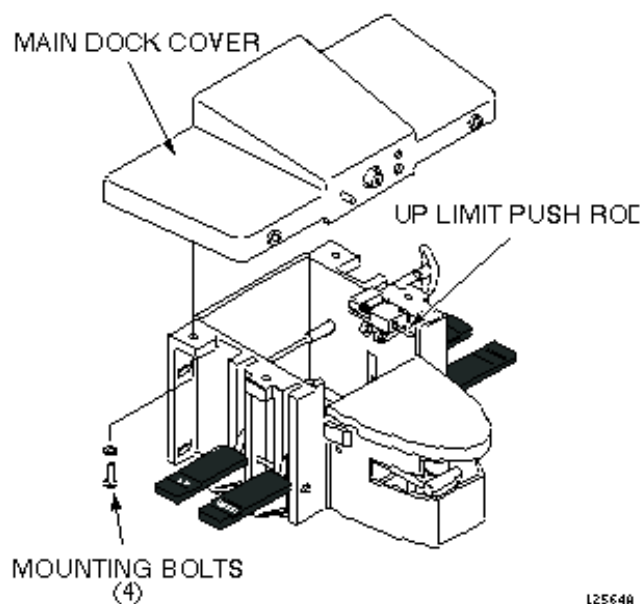
SAFETY REQUIREMENTS:

- **THE SERVICE PATH MUST ALLOW FOR THE DOCK TO BE MOVED ALONG THE GROUND A DISTANCE EQUAL TO THAT OF THE DISTANCE BETWEEN THE MAGNET AND THE FOOT-END OF THE PATIENT TABLE PRIOR TO BEING LIFTED VERTICALLY.**
- **TWO (2) MR SAFETY TRAINED PERSONNEL MUST BE PRESENT AT ALL TIMES WHEN SERVICING HIGHLY FERROUS DEVICES IN THE AREAS OF MAGNETIC FIELDS.**

WHEN PLANNING A SERVICE PATH, IT IS CRITICAL THAT THE PATH BE CLEAR AND SUFFICIENTLY WIDE. ENSURE THAT THERE ARE NO TRIP HAZARDS, OBSTACLES, CLUTTER, SLIPPERY SURFACES OR OTHER ITEMS EVEN PARTIALLY RESTRICTING THE PATH. IF THERE ARE PORTABLE OBSTACLES IN A PATH, REMOVE THEM FROM THE AREA AND REPLACE THEM AFTER THE SERVICE ACTION IS COMPLETED. IT IS REQUIRED TO WALK THE PATH PRIOR TO BEGINNING SERVICE TO ENSURE THAT THERE IS SUFFICIENT SPACE THROUGH WHICH TO PASS FOR YOURSELF AND THE OBJECT BEING SERVICED.

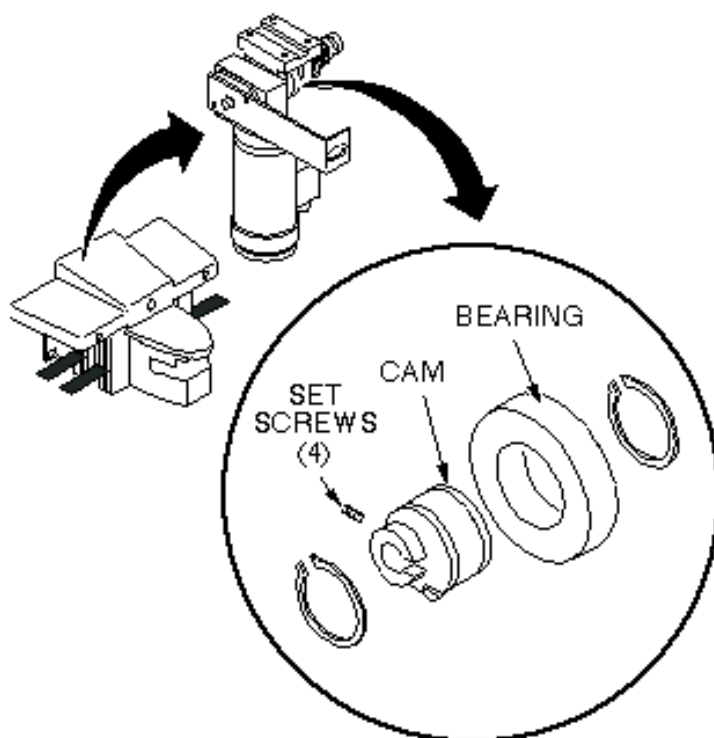
2. Remove dock from the Magnet Room following instructions in *Section 3-1-1 Dock Assembly Removal in SRI and Dock Hardware* (ME1REA1.PDF).

3. Remove the four mounting bolts (shown in Illustration 4-1) from main dock cover.



DOCK COVER REMOVAL
ILLUSTRATION 4-1

4. Slide the main dock cover forward until Up Limit Switch pushrod just clears dock cover holes. (See Illustration 4-1.)
5. Check the four set screws on motor cam shown in Illustration 4-2. If the set screws are loose, apply Loctite® 242 and Primer T to the set screws and tighten. Allow 15 minutes to harden.



DOCK MOTOR CAM ADJUSTMENT
ILLUSTRATION 4-2

6. Carefully position main dock cover so that Up Limit Switch pushrod is aligned with upper hole in dock cover. (See Illustration 4-1.)
7. Replace the main dock cover and secure with four mounting bolts shown in Illustration 4-1.

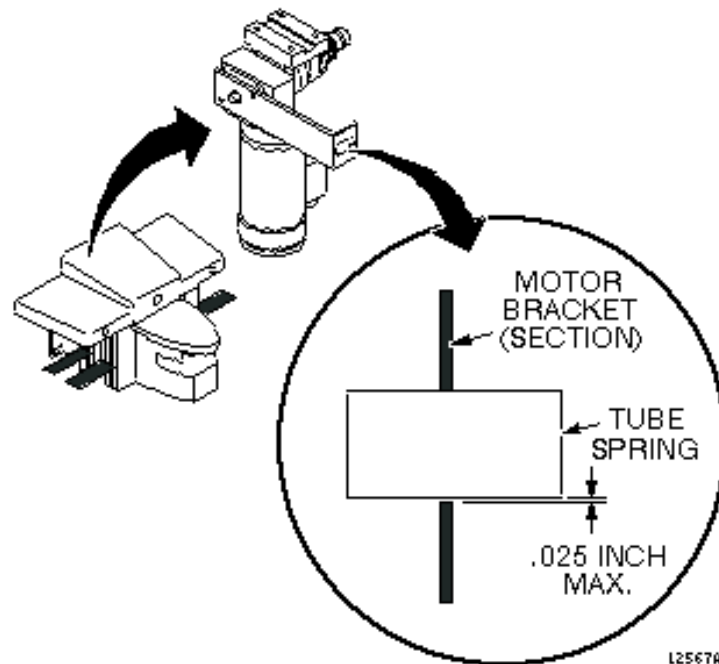


FERROUS MATERIAL HAZARD! THE DOCK ASSEMBLY MAIN DOCK COVER CONTAINS A MOTOR MADE WITH FERROUS COMPONENTS. HOLDING THIS ASSEMBLY TOO CLOSE TO THE MAGNET BORE WILL CAUSE IT TO BE FORCIBLY ATTRACTED TO THE MAGNET. TO PREVENT POSSIBLE BODILY INJURY OR DAMAGE TO MOTOR, MAGNET ENCLOSURE OR MAGNET, PLACE DOCK ASSEMBLY ON FLOOR AND SLIDE IT OUT OF EXAM ROOM.

8. Reinstall the dock in the Magnet Room following instructions in *Section 3-1-2 Dock Assembly Installation* in *SRI and Dock Hardware* (ME1REA1.PDF)
9. Exercise Table Up pedal to ensure gear noise is eliminated. If not, remove dock assembly as detailed in Section 4-1 and proceed to *Section 4-2 Motor Tube Spring Check*.

4-2 Motor Tube Spring Check

1. Remove the dock from the Magnet Room following instructions in *Section 3-1-1 Dock Assembly Removal* in *SRI and Dock Hardware* (ME1REA1.PDF).
2. Check the play between tube spring and motor mounting bracket. (See Illustration 4-4.)



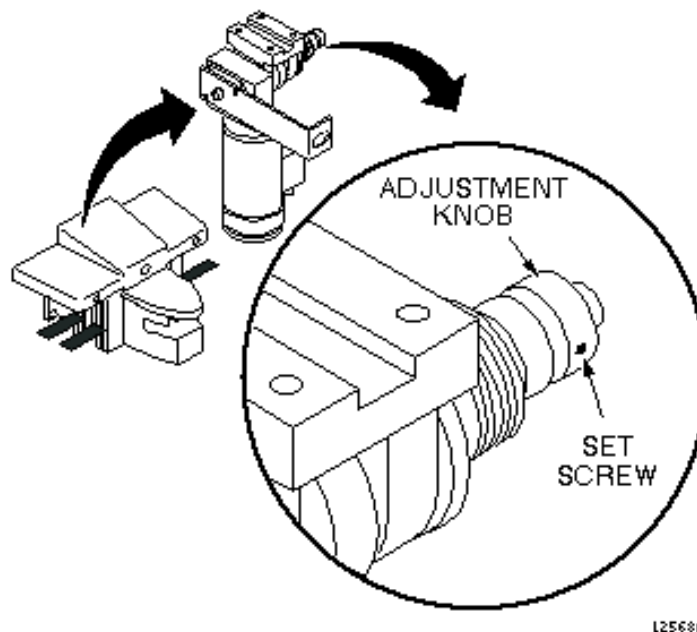
MOTOR TUBE SPRING
ILLUSTRATION 4-4

3. If play is more than 0.025 inch (0.6 mm), replace the tube spring.

4. Replace the main dock cover on dock assembly as described in Section 4-1, heeding danger precautions on previous page.
5. Reinstall the dock in the Magnet Room following instructions in *Section 3-1-2 Dock Assembly Installation* in *SRI and Dock Hardware* (ME1REA1.PDF).
6. Exercise Table Up pedal to ensure gear noise is eliminated. If not, remove dock assembly as detailed in Section 4-1 and perform *Section 4-3 Clutch Adjustment*.

4-3 Clutch Adjustment

1. Remove the dock from the Magnet Room following instructions in *Section 3-1-1 Dock Assembly Removal* in *SRI and Dock Hardware* (ME1REA1.PDF).
2. Remove main dock cover outside of the Magnet Room as outlined in Section 4-1.
3. Loosen the set screw on clutch adjustment knob. (See Illustration 4-5.)



DOCK MOTOR CLUTCH ADJUSTMENT
ILLUSTRATION 4-5

4. Adjust knob two clicks tighter (clockwise), and tighten the set screw.

Note

Do **not** adjust the clutch more than two clicks at a time. The clutch should not be tightened to the point where it affects the dock motor function.

5. Replace main dock cover on dock assembly as described in Section 4-2.
6. Reinstall the dock in the Magnet Room following instructions in *Section 3-1-2 Dock Assembly Installation* in *SRI and Dock Hardware* (ME1REA1.PDF).
7. Exercise the Table Up pedal to ensure gear noise is eliminated. If not, remove dock assembly as detailed above and repeat adjustment procedures as detailed in Sections 4-1 through 4-3.

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
A	Sept. 8, 1998	W. Pasciak	Initial release.
0	Feb. 4, 1999	W. Pasciak	Final release.
1	March 1, 1999	W. Pasciak	Added gap specification.
2	May 21, 1999	SM Atladottir	Updated Procedure References for New GUI
3	Sept 2, 1999	Resa Lambert	Moved sections around. Corrected Dock Motor Speed Adj.
4	Sept. 30, 2010	R. Kohl	Per CAPA 1859392: • Section 3-2 Motor On Switch Actuation Adjustment Procedure - Updated and added Warning. • Section 4-1 Cam Adjustment - Updated.